

Shahzaib

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Profile

Software Engineer specializing in the **MERN stack** with hands-on experience building secure scalable **RESTful APIs** and full-stack web applications. I take an **AI-first** approach to development, integrating services like the **Gemini API**, **ScrapegraphAI** and **Firecrawl** directly into production-style platforms for example building an **AI-powered shopping assistant** that scrapes and ranks live marketplace data from natural-language queries. Strong command of **JWT authentication**, **Role-Based Access Control (RBAC)**, database schema design, **Docker**-based deployments, and API documentation/testing via **Swagger** and **Postman**. A collaborative engineer comfortable in **agile Git**-driven teams focused on writing clean, maintainable code that ships reliable user-focused products.

Skills

Languages & Frameworks: JavaScript, Node.js, Express.js, React.js

Databases: MongoDB, MySQL

Backend & Architecture: RESTful API Design, JWT Authentication, Role-Based Access Control (RBAC), Database Schema, Design

AI Integration: Gemini API, ScrapegraphAI, Firecrawl, Puppeteer

Version Control System: Git, GitHub

CI/CD Tools: Docker, GitHub Actions

Additional Skills: Postman, Swagger

Professional Experience

Trainee MERN Stack Developer

Tekflix, Lahore

Mar 2026 – May 2026 (3 months)

- Engineered **RESTful APIs** in **Node.js** and **Express.js**, structuring routes, controllers, and middleware to support core application features end-to-end.
- Designed **JWT**-based authentication flows – covering token generation, validation, and protected routing – to secure user sessions across the platform.
- Built a **Role-Based Access Control (RBAC)** system to enforce endpoint-level permissions across distinct user roles.
- Modeled and optimized **MongoDB** schemas and queries to improve data retrieval performance and support efficient application-level data handling.
- Partnered with frontend engineers to define API contracts, enabling smooth integration between backend services and **React**-based interfaces.
- Containerized backend services with **Docker**, ensuring consistent environments from local development through deployment.
- Documented and tested all API endpoints with **Swagger** and **Postman**, maintaining accurate, up-to-date references for cross-team use.
- Practiced **Git**-based version control with branching strategies and code review workflows in a collaborative team setting.

Projects

(Searchify) – AI-Powered Shopping Assistant (*Final Year Project*)

Problem Solved: Traditional e-commerce search relies on rigid keywords, often returning irrelevant or overwhelming results when users don't know exact product names. Searchify addressed this by letting users describe what they want in plain language and returning ranked, trustworthy results pulled from live marketplace data.

- Built a full-stack **MERN** application (**React.js**, **Node.js**, **Express.js**, **MongoDB**) enabling users to search products using **natural language** instead of rigid keywords.
- Integrated **ScrapegraphAI** and **Firecrawl** to scrape live marketplace data in real time, layering in **AI-based ranking** to score results by review sentiment, popularity, and seller reputation.
- Designed the recommendation pipeline to surface ranked, trustworthy results directly from live data rather than a static catalog, improving relevance over traditional keyword search.

(HireMind AI) – Resume Analysis Platform

Problem Solved: Job seekers often struggle to know whether their resume truly matches a job description, where their skill gaps are, or how to prepare for interviews. HireMind AI addressed this by automating resume analysis, gap identification, and interview preparation in one platform.

- Built a **MERN** stack platform (**React.js**, **Node.js**, **Express.js**, **MongoDB**) that analyzes resumes against job descriptions, surfaces skill gaps, and generates tailored interview questions.
- Integrated the **Gemini API** to power AI-driven interview preparation and used **Puppeteer** to auto-generate downloadable, **ATS-optimized** PDF resumes.
- Implemented **JWT authentication** to secure user sessions across resume uploads, analysis history, and generated documents.

(Real-Time Collaborative Development Workspace with AI Code Assistant)

Problem Solved: Developers often rely on separate tools for team communication, project collaboration, code editing, and AI assistance, resulting in fragmented workflows. This platform consolidates these capabilities into a single workspace, enabling real-time collaboration, intelligent code assistance, and seamless project management.

- Developed a full-stack **MERN** application (**React.js**, **Node.js**, **Express.js**, **MongoDB**) featuring secure **JWT** authentication, project-based workspaces, collaborator management, and real-time messaging.
- Integrated **Socket.IO** for bidirectional real-time communication, **Redis** for session and cache management, and the **Google Gemini API** to provide AI-powered code generation, debugging assistance, and conversational support.
- Implemented an in-browser code editor using the **WebContainer API** with interactive file management, live code execution, and collaborative development environments to improve developer productivity.

(Zoomatic) – Academy Management System

Problem Solved: Academies often rely on manual, disconnected processes for managing students, course enrollment, lecture access, and scheduling. Zoomatic addressed this by centralizing these workflows into a single web platform for students and administrators.

- Designed and built a full-stack **MERN** application (**React.js**, **Node.js**, **Express.js**, **MongoDB**) to digitize academy operations – student management, course enrollment, lecture access, and scheduling – replacing manual, disconnected workflows.
- Modeled **MongoDB** schemas for student profiles, courses, lectures, enrollment records, and meeting schedules to support the platform’s core academic workflow.
- Implemented **JWT**-based authentication paired with **role-based access control** to separate student and administrator permissions across the platform.
- Built responsive interfaces in **React.js**, ensuring a consistent experience across devices for both student and administrator dashboards.

Languages

English (Professional Proficiency), Urdu (Native)